

R.A.VENUJA HIMATH RANASINGHE

Data Science Undergraduate

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PROFESSIONAL SUMMARY

Data Science undergraduate with hands-on experience developing machine learning models for risk prediction and healthcare analytics. Currently building MoodFit, a data-driven mental wellness application focused on mood tracking and behavioral insights. Proficient in Python, Scikit-learn, and XGBoost, with experience in data preprocessing, feature engineering, and model deployment using FastAPI. Passionate about developing scalable, real-world AI solutions.

KEY PROJECTS

Loan Risk Assessment System - ([Github](#))

Jan 2026 - May 2026

Developed a machine learning-based system to predict loan default risk and assist financial decision-making.

Key Features:

- Built classification models (Random Forest / XGBoost) to predict loan risk
- Implemented data preprocessing, feature selection, and model evaluation
- Used SHAP for model interpretability and explainability
- Developed a recommendation engine to suggest optimal loan terms
- Integrated backend APIs using FastAPI

Tech Stack: Python, Scikit-learn, XGBoost, SHAP, FastAPI

FitTrack – Full Stack Fitness Tracking Platform - ([Github](#))

Mar 2026 - May 2026

- Built a comprehensive fitness tracking app with features for workout/meal planning, goal setting, and user progress visualization. Conducted feasibility studies and risk assessments, mitigating potential project risks.
- Designed a scalable backend supporting 8+ core database models and automated admin initialization
- Engineered 30+ interactive screens including real-time progress dashboards using react-native-chart-kit and streak tracking

Tech Stack: React Native (Expo), Node.js (Express v5), MongoDB (Mongoose), JWT, Tailwind CSS (NativeWind)

Stroke Risk Prediction System

Jun 2025- Nov 2025

- Developed a predictive model to identify high-risk stroke patients by analyzing multifaceted health data and clinical indicators
- Implemented comprehensive data preprocessing, including feature scaling and outlier detection, to enhance model performance.
- Utilized Random Forest algorithms and evaluated results using a multi-metric approach (Precision, Recall, F1-score) to ensure diagnostic accuracy

Tech Stack: Python, Pandas, Scikit-learn, Machine Learning, Data Analytics

- Developed a data-driven application focused on mood tracking and behavioral insights using XGBoost.
- Implemented model deployment via FastAPI to provide real-time mental wellness recommendations

Tech Stack: Python, XGBoost, Scikit-learn, FastAPI, Pandas.

Vehicle Rental Management System - ([Github](#))**jan 2025 – May 2025**

- Engineered a scalable, web-based enterprise solution designed to streamline the lifecycle of vehicle rentals, from inventory tracking to customer bookings.
- Developed a robust backend architecture using Spring Boot, implementing RESTful services to handle complex business logic and high-concurrency booking requests.
- Designed and optimized a MySQL relational database schema to manage interconnected data for customers, diverse vehicle fleets, and real-time availability status.

Tech Stack: Java, Spring Boot, MySQL, Hibernate/JPA.

EDUCATION

BSc (Hons) in Data Science | [SLIIT — Sri Lanka Institute of Information Technology](#)

2024 – Present

► Actively involved in AI-related projects exploring machine learning, neural networks, and data analytics.

G.C.E. Advanced Level – Physical Science Stream

► Combined Mathematics (C), ICT (B), Physics (S)

SKILLS & TECHNOLOGIES

AI & Machine Learning

Python · ML Algorithms · Neural Networks Data Analytics · Predictive Modelling

Software Development

Java · JavaScript · Node.js · React Native Web Design · Full-Stack Development

Leadership & Management

Project Management · Team Leadership Content Strategy · UX Design

ACHIEVEMENTS & HIGHLIGHTS

- Champion – Colombo Zonal, National Youth Blitz Chess Championship (2019)
- 1st Runner Up – Colombo Zonal, National Youth Blitz Chess Championship (2014)
- 9th Place – Sri Lanka Novices Chess Championship
- Captained the school chess team, leading strategy and team coordination